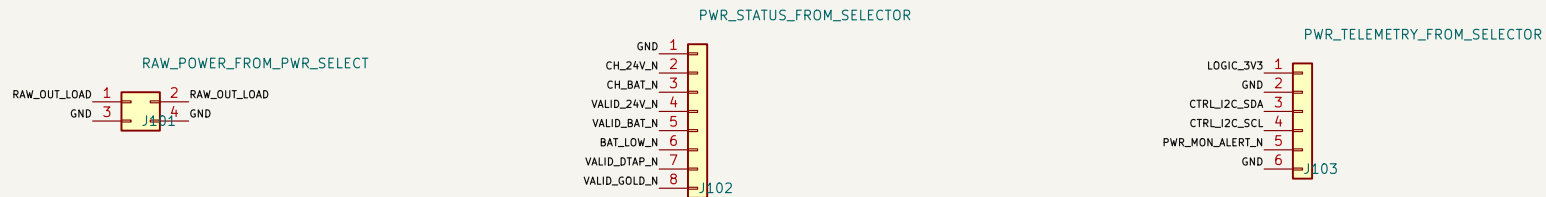


A. PWR-SELECT INTERFACE



Mates with PWR-SELECT J301; two contacts per polarity validate 15 A derating.

Mates with PWR-SELECT J401; add 3.3 V pull-ups on the carrier.

Mates with PWR-SELECT J402; carrier supplies 3.3 V and owns I2C pull-ups.

B. BUFFERED DIFFERENTIAL TDM + AUDIO CONTROL

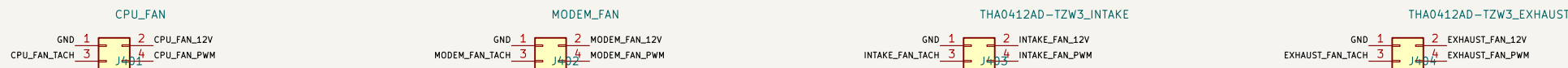


Separate filtered audio-power harness; no fan/radio loads.

Mates with AUDIO-BXB J101 using Molex 87832-6423 headers; harness assembly remains vibration-qualified.

Carrier drives MCLK/BCLK/FSYNC/DAC data; AUDIO-BXB drives ADC data back.

C. FOUR INDEPENDENT PWM / TACH FANS



J403 is physically keyed/labelled INTAKE; J404 is keyed/labelled EXHAUST.

D. DETAILED CAPTURE SUITE – REV A1

01 CMS-Core-Allocated; exact 300-contact CMS symbol, 76 locked functions

02 CMS-Core-Allocated; VCC, SYSIN, reset, recovery, power-on and debug UART

03 Network-PCIe: PITC9X256086P, 3x LAN7430 and native WAN1 front end

04 Network-PCIe: Nx Wurth MagJack and Molex 0679101002 Mini PCIe 4TAR Wi-Fi

05 WWAN-SIM: TE 2199230-3 B-key, USB3/USB2, controls and dual-SIM mux

06 Display-Harness: HDMI, USB touch and dedicated 12 V / 2.5 A branch

07 Audio-Control: I2SO TDM, I2S1 ES8316 headset and CTIA amplifier path

08 Thermal-ID: I2C translation, GPIO expansion, sensors and four fan channels

09 Power-Regulators-A1: protected raw hold-up, all main rails, sequencing and test points

10 Audio-Bxb: all XLR shields bonded to chassis with one controlled AGND bond

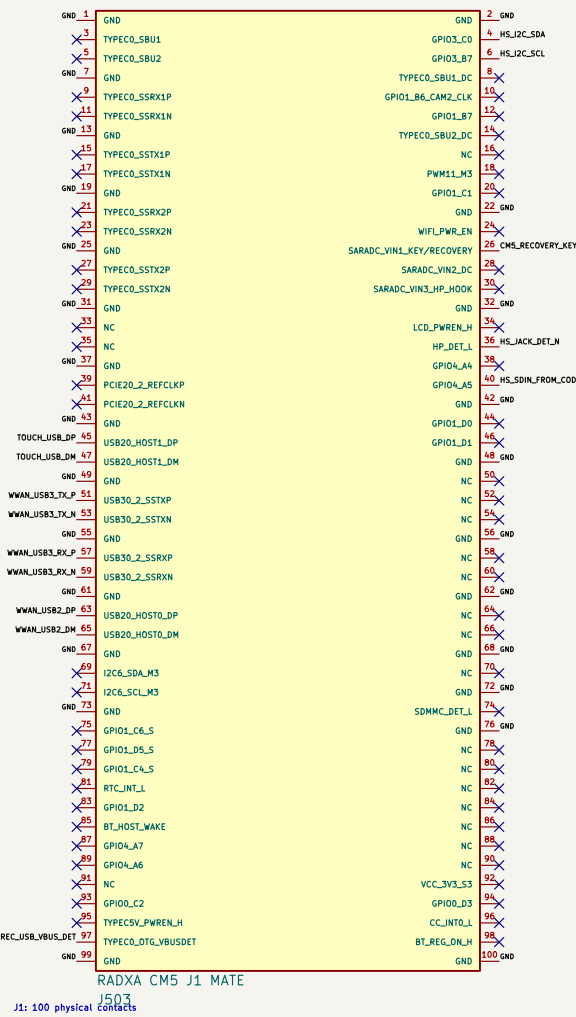
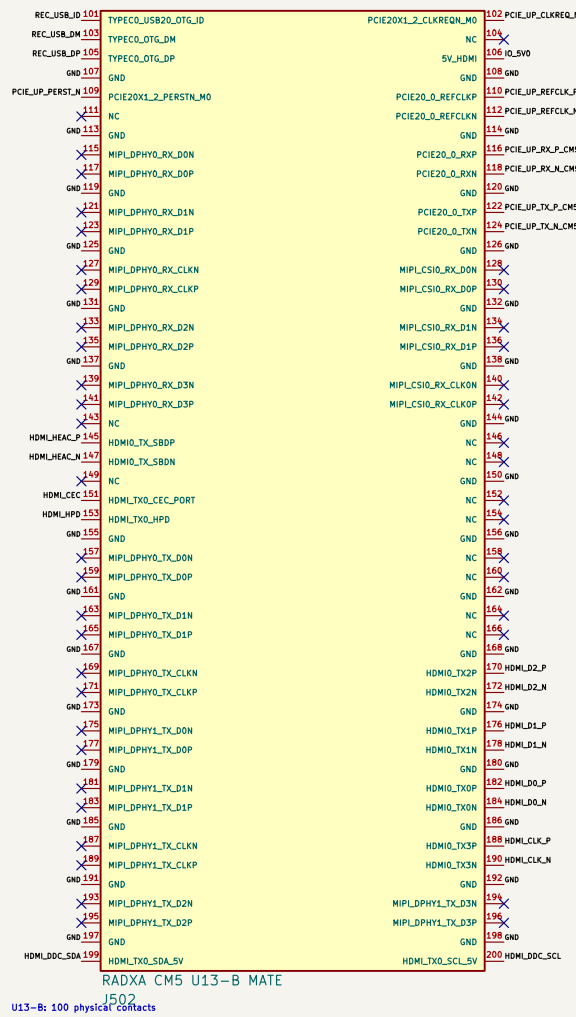
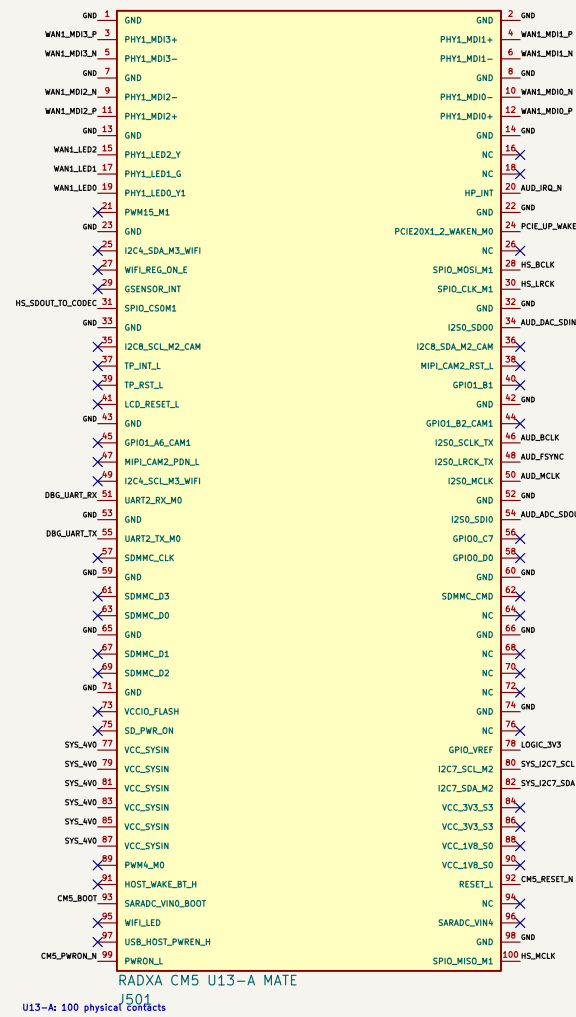
11 Review: every detailed sheet has zero ERC errors; warnings are classified

DC/DC compensation and magnetics remain a separate power-design calculation milestone.

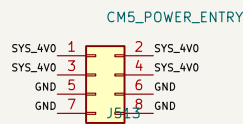
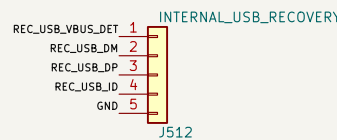
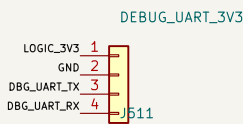
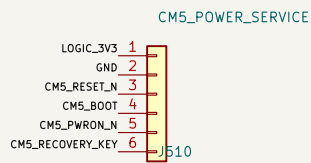
High-speed routing starts only after stackup and connector 2 datums are released.

The two enclosure fans require an underside baffle to prevent direct intake/exhaust short flow.

1. EXACT RADXA CM5 MATING CONNECTORS



2. INTERNAL POWER / STARTUP / RECOVERY SERVICE



3.3 V UART only; keyed factory/debug cable.

Internal keyed recovery harness; service fixture owns the protected USB-C receptacle and CC network.

Kelvin-sense SYS_AVO at J501; 4.0 V follows the Radxa CM5 carrier design note.

Internal keyed service header; never exposed on the top panel.

3. OWNERSHIP AUDIT

76 / 76 allocated contacts connected; 0 duplicate physical-pin claims.

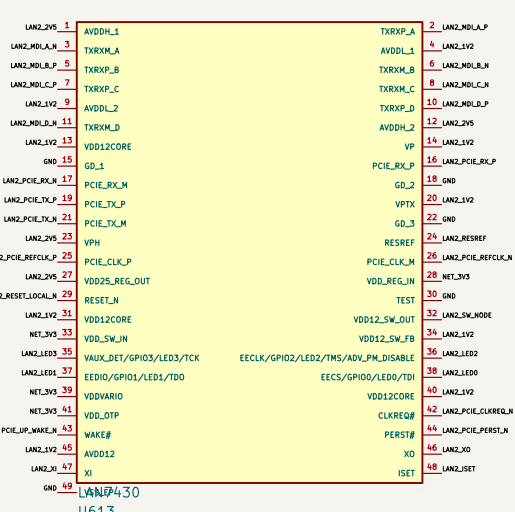
WAN1 uses native MDL PCIe Gen2 x1 feeds the packet switch.

WAN owns USB30_2 + USB20_HOST0; touch owns USB20_HOST1.

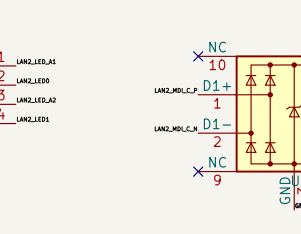
HDMI_TX0, I2S0 TDM, I2S1 headset, I2C7, I2C3 and UART2 are reserved exactly once.

Pins 145/147 remain HDMI HEAC reserved. U13-B pin 106 receives the dedicated IO_5V0 rail.

Unused alternate-function contacts carry explicit no-connect markers on this A1 baseline.



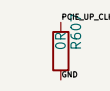
74991114412



74991114412

74991114412

74991114412



VAR: 7109111W2 voltage-mode 5 Gb/s vicinals direct to class's, PFI-side residual 1



GP90[1:0] = 01 selects 606 mode. GP901 uses its internal pulldown; GP900 is pulled high.

Hold PERST# low until 3.3 V, 1.0 V, and 100 MHz REFCLK are stable for at least 100 ns.



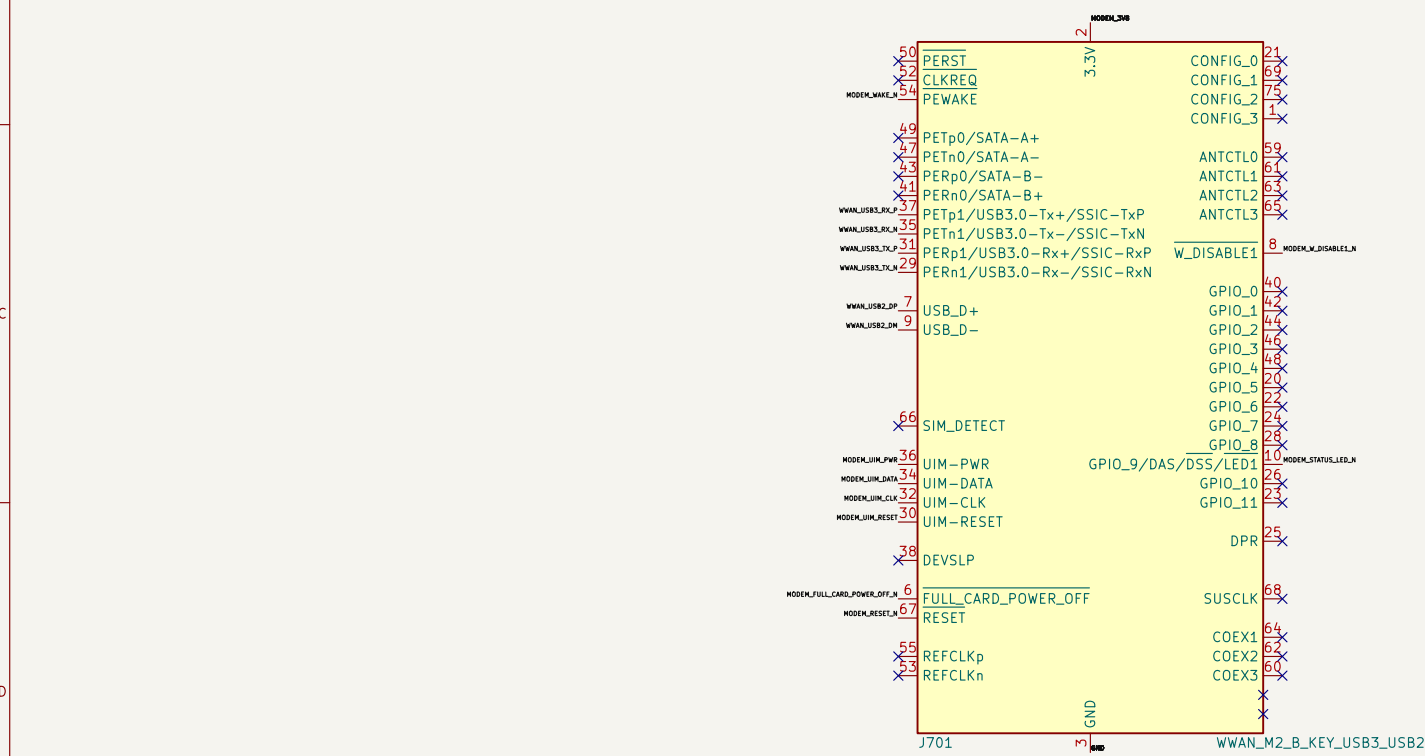
ZK7915-8P1: four WS-FE 6. ATVR 80-d PCIe AP, 3.3 V / 4 A rail, four RF slots

J629 uses the controlled 58-67960-006 C2 land pattern; 38/first-article fit remains a release gate.

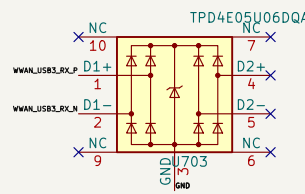
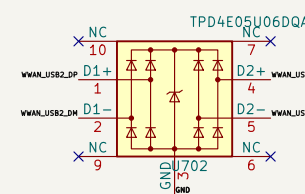


PWR_FLAG symbols declare off-sheet regulator outputs for ERC; they are not physical parts.

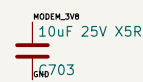
1. UNIVERSAL M.2 B–KEY WWAN SOCKET



2. USB ESD AND MODEM BULK CAPACITANCE



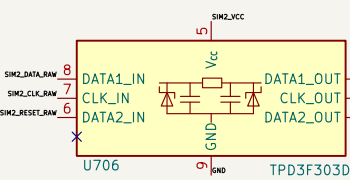
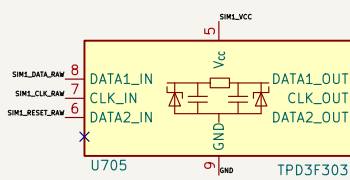
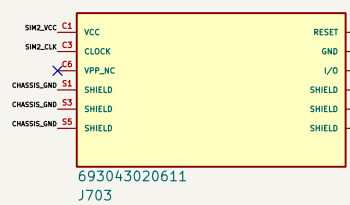
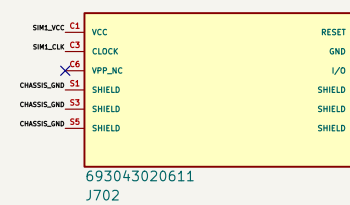
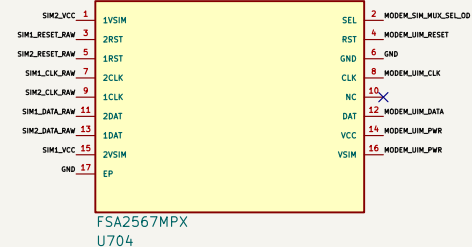
Place USB ESD at the modem socket: route USB 2 pairs as short, continuous 90 ohm differential channels.



U702 keeps SCL high SIM 1 is selected by default; SPD control must be open drain; low selects SIM 2.

The control domain blocks MODM_MUX_PMR, providing a fixed 3.3 V condition when the SIM rail is 1.8 V.

PHY004 uses TXCDT channel 2 and physical SIM 2 uses channel 1 to implement that default.



EN0: TPD3F3030 RESET/CLK/EN0 and channel VCC inside the modem shell leads to CMDSIGNAL.

EN0: TPD3F3030 RESET/CLK/EN0 and channel VCC inside the modem shell leads to CMDSIGNAL.

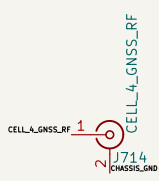


Power-Regulator (L189-L192) provide L20 of low-ESR bulk inside the socket; CTR5/CTR6 are local RF bypass.

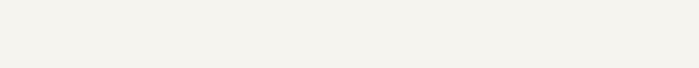
6. MODEM CONTROL CONTRACT



5. RF BULKHEAD HARNESS CONTRACT



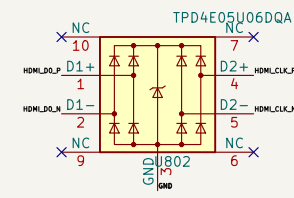
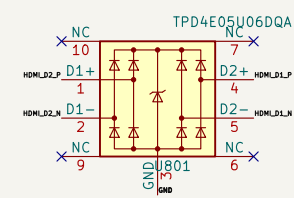
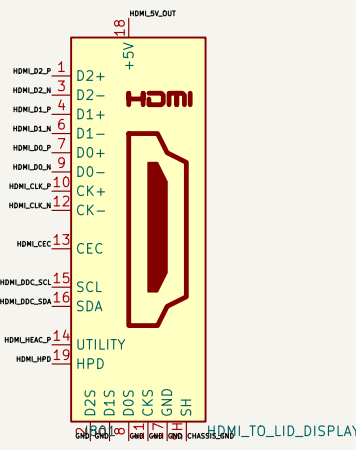
Four left-based EXT R048333M uses RF 10-to-10M bulkhead pigtail; their cable length requires a model sample gain.



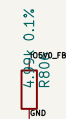
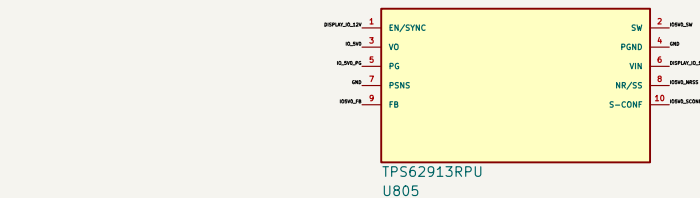
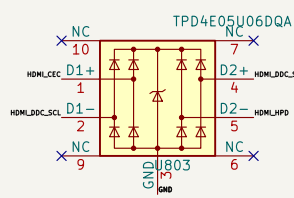
1. HDMI TYPE-A SOURCE CONNECTOR

2. HDMI ESD / LOW-CAPACITANCE SHUNTS

3. DEDICATED IO_5V0 / 2 A BUCK



Capacitor CM5 005 5.1 tolerance and their pull-up values against the Pinmux center reference before layout review.



HDMI_HDMI_0 connects exposed to Row A, the attached monitor uses video/D0+/D0-/D1+/D1-/D2+/D2-/D3+/D3-/D4+/D4-/D5+/D5-/D6+/D6-/D7+/D7-/D8+/D8-/D9+/D9-/D10+/D10-/D11+/D11-/D12+/D12-/D13+/D13-/D14+/D14-/D15+/D15-/D16+/D16-/D17+/D17-/D18+/D18-/D19+/D19-.

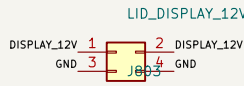
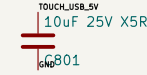
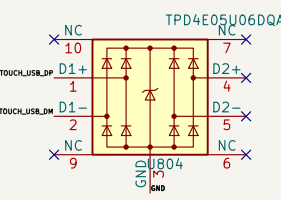
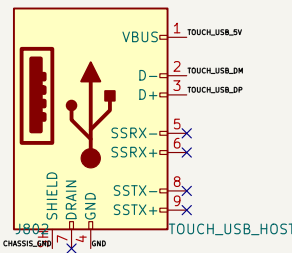
Start bench to check at the connector. TMSD signal returns use the local right ground plane.

Place all HDMI ESD arrays immediately behind (B0) with no holes on TMSD pins.

ESD array CM5 005 5.1 tolerance and their pull-up values against the Pinmux center reference before layout review.

4. USB TOUCH HOST PORT

5. 12 V / 2.5 A DISPLAY POWER



USB 2.0 is allocated from the CM5. A standard USB 2.0-to-4 cable with carrier USB 2.0 touch data.

The monitor-side SuperSpeed Type-B receptacle is not illustrated as a device connector on this host carrier.

ESD array CM5 005 5.1 tolerance and their pull-up values against the Pinmux center reference before layout review.

No local display offset. Use the upstream B0_12V branch fuse and B0_12V input 10 AHS harness.

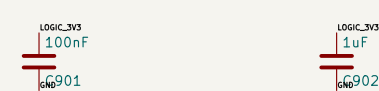
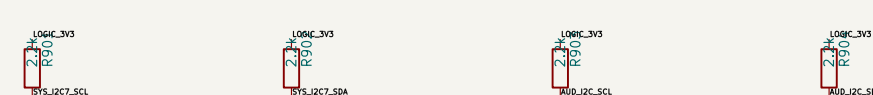
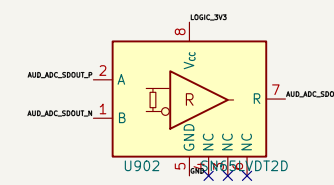
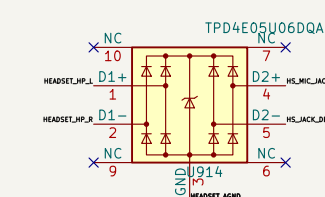
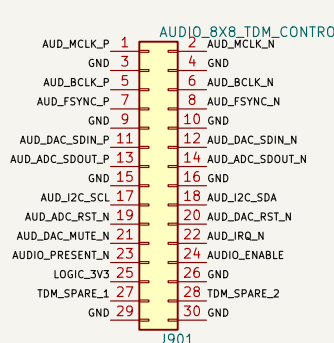
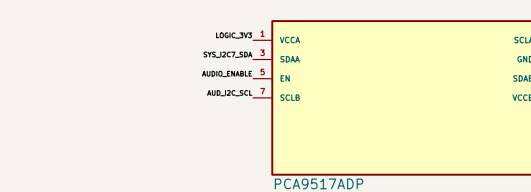
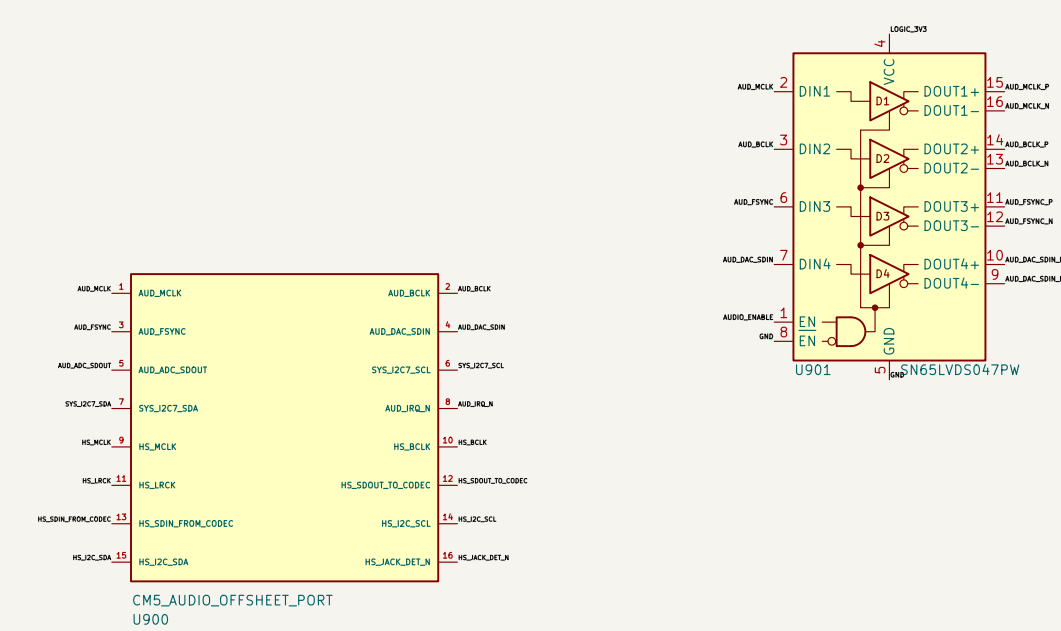
6. HINGE-EDGE HARNESS BUILD

Radxa CM5 USB touch, B0_12V 12 to 18V +/- 25 mm each, with separate related hinge and service loops.

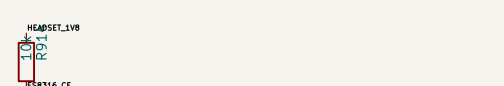
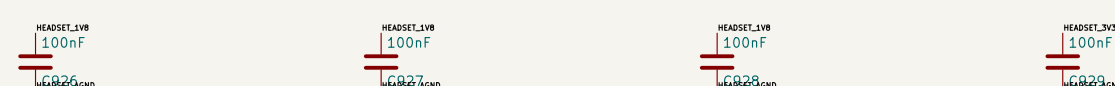
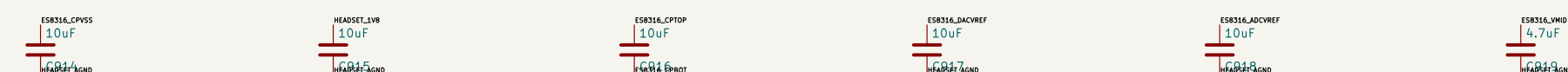
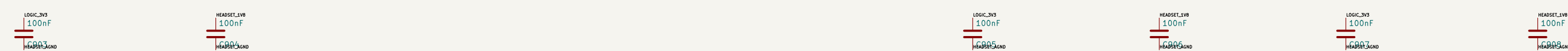
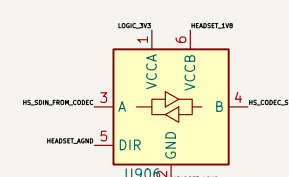
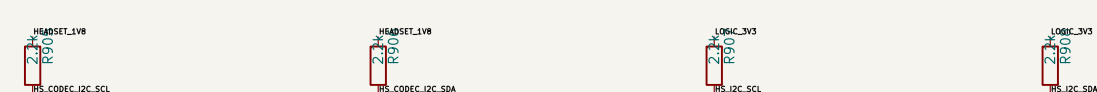
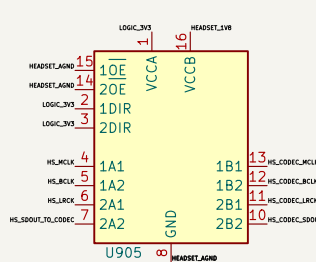
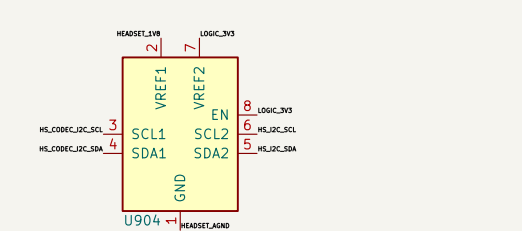
Assemblies: full 10 board plus 100 mm panel 10 / 10 degree 100 to connector flexion at hinge-free bend.

Carrier connectors face down. Show related loop area of loop, P02 offset, board thickness and chip edges.

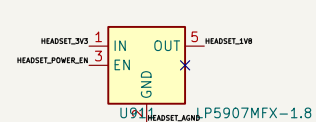
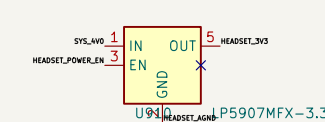
1. I2S0/TDM PROGRAM-AUDIO LVDS LINK



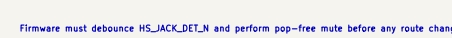
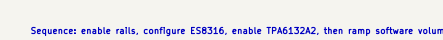
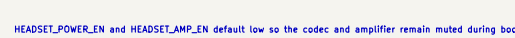
2. I2S1 HEADSET LEVEL TRANSLATION



3. DEDICATED LOW-NOISE HEADSET RAILS



7. HEADSET CONTROL POLICY

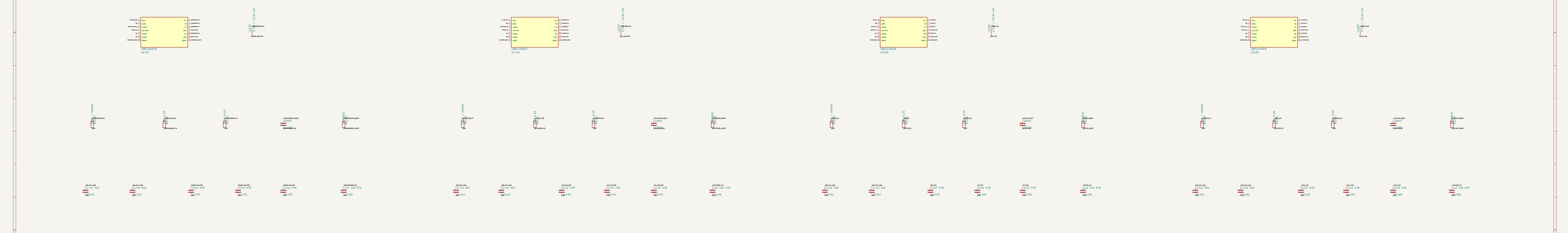


1. PROTECTED RAW INPUT FROM PMR-SELECT

2. SYS_4V0 / 12 A SYNCHRONOUS BUCK

3. AUX_12V / 8 A FOUR-SWITCH BUCK-BOOST

4. DEDICATED RADIO, NETWORK, AND LOGIC BUCKS



5. LOW-NOISE POINT-OF-LOAD RAILS

6. AUX_12V BRANCHES AND AUDIO HANDOFF

